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Topology Advanced Class MT24 : Factorization Algebras

Week 3

Free Field Theories

Ref. (Costello-William) Factorization Algebras in Quantum Field Theory Vol 1 Ch (2,4, Appendix A)
Vol 2 ch(2.5, 6,8)

§1. Definition of a classical field theory

§2. Introducing Po-operad and BD-operad

§3. Classical observables

§4. Deformation quantization - Quantum observables

§5. Correlation functions

§6. Example - (massive) free scalar theory, Yang-Mills theory.

§1. A succinct definition of a classical field theory.

Defn 1.1 Let V be a vector space of dim d . A **density** σ on V is a function $\sigma : \Lambda^d V \rightarrow \mathbb{R}$ such that $\sigma(xv) = |x| \sigma(v) \quad \forall x \in \mathbb{R} \text{ and } v \in \Lambda^d V$.

Denote 1-dim vector space of densities on V by Dens_V .

Defn 1.2 The **density line bundle** on a manifold Dens_M is a smooth bundle whose fiber over $x \in M$ is the 1-dimensional space $\text{Dens}_{T_x M}$ of densities on $T_x M$.

Sections of Dens_M on open set U are smooth densities on U . These can be integrated.

Densities are differential geometric analogues of measure.

Defn 1.3 Let E be a smooth vector bundle over M . The **∞ -jet bundle** $\text{Jet}(E)$ is the vector bundle whose fiber at $x \in M$ is the space of jets (formal germs at x of sections of E). We denote the sheaf of sections of $\text{Jet}(E)$ as $\mathcal{J}(E)$.

Defn 1.4 For two vector bundles E_1 and E_2 over M , let $\mathcal{E}_1$ and $\mathcal{E}_2$ denote the sheaf of sections respectively. A **differential operator** $D: \mathcal{E}_1 \rightarrow \mathcal{E}_2$ is map of sheaves that factors as

$$\begin{array}{ccc} \mathcal{E}_1 & \xrightarrow{D} & \mathcal{E}_2 \\ \text{'jet prolongation of a section'}: \mathcal{J} \searrow & & \nearrow \tilde{D} \\ & \mathcal{J}(E_1) & \end{array} \quad \text{where } \tilde{D} \text{ is induced by a map of vector bundles } \text{Jet}(E_1) \rightarrow E_2.$$

Defn 1.5 Let E be a graded vector bundle on M . Denote $E^! = E^\vee \otimes \text{Dens}_M$.

A (-1) -shifted symplectic structure on E is an isomorphism of graded vector bundles

$$E \simeq E^![-1]$$

which induces an anti-symmetric pairing $\langle \cdot, \cdot \rangle_{\text{loc}} : E \otimes E \rightarrow \text{Dens}_M[-1]$.

$$\langle \xi_1, \xi_2 \rangle_{\text{loc}} = -(-1)^{|\xi_1||\xi_2|} \langle \xi_2, \xi_1 \rangle$$

$\mathcal{E}_c$ is the sheaf of compactly supported sections of E .

for each open $U \subset M$, there is a (-1) -shifted anti-symmetric pairing

$$\langle \cdot, \cdot \rangle : \mathcal{E}_c(U) \otimes \mathcal{E}_c(U) \rightarrow \text{Dens}_M(U)[-1] \xrightarrow{\int_U} \mathbb{R}[-1].$$

i.e. for $\phi, \psi \in \mathcal{E}_c(U)$

$$\langle \phi, \psi \rangle = \int_{x \in U} \langle \phi(x), \psi(x) \rangle_{\text{loc}}$$

$|\phi| + |\psi| = 1$ for non-trivial pairing.

Physics : $\mathcal{E}(U) \sim$ space of fields of a theory (arising as sections of a vector bundle)

A pre classical free field theory is the following data:

- (i) a manifold M (spacetime)
- (ii) A $\mathbb{Z}$ -graded vector bundle E on M with a (-1) -shifted symplectic structure.
- (iii) A differential operator on the sheaf of sections of E

$$Q: \mathcal{E} \rightarrow \mathcal{E}$$

of cohomological degree $+1$, square zero and skew-self adjoint
with respect to $\langle, \rangle$ on $\mathcal{E}_c$.

$$\langle \varphi, Q\psi \rangle = -(-1)^{|\varphi|} \langle Q\varphi, \psi \rangle.$$

for φ, ψ of homogeneous
degree in $\mathcal{E}_c$.

A pre classical free field theory is a classical if $(\mathcal{E}, Q)$ is
an elliptic complex.

An elliptic complex is a $\mathbb{Z}$ -graded vector bundle and a differential operator $Q^n: \mathcal{E}^n \rightarrow \mathcal{E}^{n+1}$ s.t. $Q^2 = 0$
and whose principal symbol complex is exact on $T^*M \setminus M$

Thm. The cohomology groups of an elliptic complex on a closed manifold are
finite dimensional.

Recovering the built-in physics of free field theories

for $\phi \in \mathcal{E}_c(U)$ for $U \subset M$

$S(\phi) := \langle \phi, Q \phi \rangle$ is the action.

Physics : For a field theory we find fields that extremise S (called physical fields)
critical locus $\text{crit}(S)$

i.e we want to solve $dS=0$



Theorem (C-G): A free field theory $(M, \mathcal{E}, Q, \langle, \rangle)$ has

$\text{Obs}^{\mathcal{C}}$

(i) a canonical P_0 -prefactorization algebra of classical observables.

$\text{Obs}^{\mathcal{Q}}$

(ii) a canonical BD-prefactorization algebra of quantum observables.

§2. P_0 -operad and BD-operad.

Defn 2.1 The P_0 -operad is the graded operad generated by a commutative associative product $--$ of degree 0 and a Poisson bracket of degree 1.

A P_0 -algebra is a cdga with a 1-shifted Poisson bracket.

A pre factorization algebra is a P_0 (or 1-shifted Poisson) pre factorization algebra if

- (i) each $F(U)$ is a P_0 -algebra
- (ii) the corestriction maps are maps of P_0 -algebras
- (iii) The sections of disjoint opens U_i under the factorization maps

$$F(U_1) \otimes \dots \otimes F(U_n) \rightarrow F(V) \text{ Poisson commute in } F(V).$$

"It is pre-factorisation algebra valued in $P_0\text{-Alg}(\mathcal{C})$ for some category $\mathcal{C}$."

Defn 2.2 The E_0 -operad is the operad with $E_0(0) = \mathbb{R}$ and $E_0(n) = 0 \quad \forall n > 0$.

An E_0 -algebra is a pointed object.

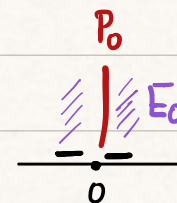
Defn 2.3 The BD operad is an operad in dg modules over $\mathbb{R}[[\hbar]]$ where $\hbar$ has cohomological degree 0, whose underlying graded operad agrees with the P_0 operad (tensored up to $\mathbb{R}[[\hbar]]$), i.e. $BD = P_0 \otimes_{\mathbb{R}} \mathbb{R}[[\hbar]]$ equipped with the differential that satisfies $d(- \cdot -) = \hbar \{ , \}$ on $BD(2)$ and determines the differential on the n -ary operations.

Beilinson-Drumfeld 2004

A BD -algebra structure on a flat dg $\mathbb{R}[[\hbar]]$ -module M is

- (1) a commutative associative product $- \cdot -$ of degree 0.
- (2) a Poisson bracket $\{ , \}$ of degree 1, the internal differential $d = d_M$ being a derivation for this bracket.
- (3) $d(m \cdot n) = (d m) \cdot n + (-1)^{|m|} m \cdot (d n) + \hbar \{ m, n \}$ holds.

Thm (i) $BD \otimes_{\mathbb{R}[[\hbar]]} \mathbb{R} \cong P_0$ an iso of operads.



"interpolates between P_0, E_0 ".

(ii) $BD \otimes_{\mathbb{R}[[\hbar]]} \mathbb{R}((\hbar)) \cong E_0 \otimes \mathbb{R}((\hbar))$ a quasi-iso of operads over $\mathbb{R}((\hbar))$.

Defn. A **BD quantization** of a P_0 prefactorization algebra F is a BD pre factorization algebra $F_\hbar$ and a quasi-isomorphism

$$F_\hbar \otimes_{\mathbb{R}[[\hbar]]} \mathbb{R} \cong F \text{ of } P_0 \text{ factorization algebras.}$$

§3. Classical Observables

For a free field theory $(M, \mathcal{E}, Q, \langle, \rangle)$

Intuitively observables are functions on the physical fields.

$$\text{Obs}^{\text{cl}}(U) \longleftrightarrow \text{Sym}(\mathcal{E}(U)^{\vee})$$

"distributional linear observable"

instead we use

$$\text{Sym}(\mathcal{E}_c^!(U))$$

"smoothed linear observable"

δx

↓ replaced

bump function
supported near x

Lemma: $\mathcal{E}_c^!(U)$ is chain homotopic to $\mathcal{E}(U)^{\vee}$.

Defn. $\text{Obs}^{\text{cl}}(U) = \text{Sym}(\mathcal{E}_c^!(U)) = \text{Sym}(\mathcal{E}_c(U)[1])$

$U \longmapsto$ a cdga $\text{Obs}^{\text{cl}}(U)$

$U \subset V \longmapsto i_V^U : \text{Obs}^{\text{cl}}(U) \rightarrow \text{Obs}^{\text{cl}}(V)$

"extension by zero"

disjoint

$U_1, \dots, U_n$

$\longmapsto \text{Obs}^{\text{cl}}(U_1) \times \dots \times \text{Obs}^{\text{cl}}(U_n) \xrightarrow{\text{multi-linear}} \text{Obs}^{\text{cl}}(V)$
 $(d_1, \dots, d_n) \mapsto \prod_{i=1}^n i_V^{U_i}(d_i).$

$\text{Obs}^{\text{cl}}(U)$
form a
pre-factorization
algebra.

Excursion on derivations.

Let $\mathcal{C}$ be a nice $(\infty-)$ category and let A be a commutative algebra object in $\mathcal{C}$, M be an A -module.

$\text{Der}(A, M)$ is an A -module in $\mathcal{C}$.

for $X \in \text{obj } \mathcal{C}$ and $A \in \text{cAlg}(\mathcal{C})$

$$\text{cAlg}(\mathcal{C}) \xleftarrow{\text{Sym}} \mathcal{C} \xrightarrow{U}$$

$$\text{Sym} \dashv U.$$

$$\underline{\text{Hom}}_{\mathcal{C}}(X, UA) \simeq \underline{\text{Hom}}_{\text{cAlg}(\mathcal{C})}(\text{Sym}(X), A)$$

Claim

$$\text{Der}(\text{Sym}(X), M) \simeq \underline{\text{Hom}}(X, M)$$

We consider $\text{Der}(\text{Sym}(X), \text{Der}(\text{Sym}(X), \text{Sym}(X)))$

$$\simeq \underline{\text{Hom}}(X, \underline{\text{Hom}}(X, \text{Sym}(X)))$$

$$\simeq \underline{\text{Hom}}(X \otimes X, \text{Sym}(X))$$

This is space of bi-derivations on $\text{Sym}(X)$ valued in $\text{Sym}(X)$.

$\text{Der}(A, M)$

$$\xrightarrow{\underline{\text{Hom}}_{\text{cAlg}(\mathcal{C})}} (A, A \oplus M)$$

with square zero algebra structure.

$$\xrightarrow{\underline{\text{Hom}}_{\text{cAlg}(\mathcal{C})}} (A, A \oplus M) \otimes I$$

$$\downarrow \text{id} \otimes \text{br}_A$$

$$\xrightarrow{\underline{\text{Hom}}_{\text{cAlg}(\mathcal{C})}} (A, A \oplus M) \otimes \underline{\text{Hom}}_{\text{cAlg}(\mathcal{C})}(A \oplus M, A)$$

$$\downarrow 0$$

I

$$\xrightarrow{\text{id}_A}$$

$$\underline{\text{Hom}}_{\text{cAlg}(\mathcal{C})}(A, A)$$

useful.

To give Po-algebra structure to $\text{Obs}^{\text{cl}}(U)$

shift the pairing $\langle -, - \rangle : \mathcal{E}_c(U) \otimes \mathcal{E}_c(U) \rightarrow \text{Dens}[-1] \rightarrow \mathbb{R}[-1]$
 we obtain a $(+1)$ -shifted antisymmetric pairing

$$\{, \} : \mathcal{E}_c(U)[1] \otimes \mathcal{E}_c(U)[1] \rightarrow \mathbb{R}[1].$$

Lemma:

The space of biderivations on $\text{Sym}(\mathcal{E}_c(U)[1])$ valued in $\text{Sym}(\mathcal{E}_c(U)[1])$

$$\simeq \underline{\text{Hom}}(\mathcal{E}_c(U)[1] \otimes \mathcal{E}_c(U)[1], \text{Sym}(\mathcal{E}_c(U)[1]))$$

Use the $\{, \}$ pairing and include $\mathbb{R}[1] = \text{Sym}^0(\mathcal{E}_c(U)[1])$

$$\downarrow$$

$$\text{Sym}(\mathcal{E}_c(U)[1])$$

which gives a biderivation.

Check it satisfies Jacobi:

* for disjoint $U_1, U_2 \subset V$, $\alpha_i \in \text{Obs}^{\text{cl}}(U_i)$

$$\{i_V^{U_1} \alpha_1, i_V^{U_2} \alpha_2\} = 0$$

makes Obs^{cl} a Po (pre) factorization algebra.

§4. Quantum Observables

Defn 4.1 Let $\text{Heis}(U)$ be defined as the central extension

$$0 \rightarrow \mathbb{R} \cdot \hbar [-1] \rightarrow \text{Heis}(U) \rightarrow \mathfrak{e}_c^!(U) [-1] \rightarrow 0$$

with Lie bracket

$$[\phi + \hbar a, \psi + \hbar b] = \hbar \langle \phi, \psi \rangle \quad \text{for } \phi, \psi \in \mathfrak{e}_c^!(U)$$

Defn 4.2 Chevalley - Eilenberg complex.

Given a $\mathfrak{g} \in \text{dgl a}$, consider its (homological) Chevalley - Eilenberg complex

$$CE_*(\mathfrak{g}) = (\text{Sym}^*(\mathfrak{g}[1]), D)$$

The differential D sends the product of homogeneous elements x_i in degree $|x_i|$ to

$$D(x_1, \dots, x_n) = \sum_{1 \leq i \leq n} (-1)^{|x_1| + |x_2| + \dots + |x_{i-1}|} x_1 \dots x_{i-1} d x_i x_{i+1} \dots x_n + \sum_{1 \leq i < j \leq n} (-1)^{|x_i|(|x_{i+1}| + \dots + |x_{j-1}|)} x_1 \dots x_{i-1} x_{i+1} \dots x_{j-1} [x_i, x_j] x_{j+1} \dots x_n$$

$$\text{Obs}^q(U) := CE_*(\text{Heis}(U)) = \text{Sym}(\text{Heis}(U)[1], D^q)$$

As a graded vector space

$$\begin{aligned}\mathrm{Sym}(\mathrm{Heis}(U)[\hbar]) &= \mathrm{Sym}(\mathbb{R}\hbar \oplus \mathcal{E}_c^!(U)) = \mathrm{Sym}(\mathcal{E}_c^!(U))[\hbar] \\ &= \mathrm{Obs}^c(U)[\hbar]\end{aligned}$$

The quantum observables supported in U have

- (i) product structure of degree 0 as a symmetric algebra
- (ii) Poisson bracket of degree 1 extended as a derivation from Lie bracket on $\mathrm{Heis}(U)$.

Since $\mathbb{R}[\hbar][\hbar^{-1}]$ is central, poisson bracket reduces modulo $\hbar$ to poisson bracket on $\mathrm{Obs}^c(U)$.

BD axiom follows from definition of C^*

§5. Correlation functions

Let $(M, \mathcal{E}, Q, \langle, \rangle)$ be a field theory.

and Suppose $H^*(\mathcal{E}(M), Q) = 0$

$\mathcal{E}(M)$ is the tangent complex for the formal moduli space of solutions to the equations of motion at the base point around which we are doing perturbation.

The statement that this tangent complex has no cohomology means trivial solution of eq. of motion has no deformation (isolated point in the moduli space).

Lemma 5.1 $H^*(\text{Obs}^q(M)) = \mathbb{R}[[\hbar]]$

Defn 5.2 $U_1, U_2, \dots, U_n \subset M$ disjoint open sets

$$\theta_i \in \text{Obs}^q(U_i).$$

Define $\langle \theta_1, \dots, \theta_n \rangle \in \mathbb{R}[[\hbar]]$ to be the image of product

$\theta_1 \dots \theta_n \in H^*(\text{Obs}^q(M))$ under canonical iso between
 $H^*(\text{Obs}^q(M))$ and $\mathbb{R}[[\hbar]]$

8 Examples

Free scalar field theory of manifold M with mass m
 where $V dS = \Delta + m^2$

$$1. \mathcal{E}(U) = C^\infty(U) \xrightarrow{V dS} C^\infty(U)[-1]$$

pairing on $\mathcal{E}_c(U)$

$$\langle \phi^0, \phi^1 \rangle = \int_U \phi^0 \phi^1 \text{ for } \phi^k \in C^\infty(U)[-k]$$

$$\text{obs}^0(U) = \text{Sym}(\mathcal{E}_c(U)[1]) = \text{Sym}(C_c^\infty(U)[1]) \xrightarrow{\Delta + m^2} C_c^\infty(U)$$

$$\dots \rightarrow \underbrace{\wedge^2 C_c^\infty(U) \otimes \text{Sym} C_c^\infty(U)}_{\text{space of polynomial sections of } \wedge^2 T_c C^\infty(U)} \xrightarrow{V dS} C_c^\infty(U) \otimes \text{Sym} C_c^\infty(U) \rightarrow \text{Sym} C_c^\infty(U)$$

def: $PV_c(C^\infty(U))$ polynomial vector fields on $C^\infty(U)$ along the foliation given by $T_c C^\infty(U)$

differential is contraction by $V dS$

$$\text{obs}^1(U) = \dots \xrightarrow{V dS + \hbar \text{Div}} C_c^\infty(U) \otimes \text{Sym} C_c^\infty(U) \xrightarrow{V dS + \hbar \text{Div}} \text{Sym} C_c^\infty(U)$$

CGVol - Ch 4.

2. Abelian Yang-mills (with gauge group $\mathbb{R}$)

M is 4 dimensional manifold,
 $A \in \Omega^1(M)$ is a connection on trivial $\mathbb{R}$ -bundle on M

$$S_{YM} = -\frac{1}{2} \int_M dA \wedge * dA = \frac{1}{2} \int_M A (d + d^*) A$$

equations of motion $d + d^* A = 0$

gauge symmetry $x \in \Omega^0(M) \quad A \mapsto A + dx$

$$\mathcal{E} = \Omega^0(M)[1] \xrightarrow{d} \Omega^1(M) \xrightarrow{d + d^*} \Omega^2(M)[-1] \xrightarrow{d} \Omega^3(M)[-2]$$

$H^0(\mathcal{E})$ is space of $A \in \Omega^1(M)$ that satisfy YM $d + d^* A = 0$ modulo gauge symmetry.

